equal to 1-year LIBOR − 25 bps for the remaining two years of the swap. Should Swaps 'R Us exercise its option and pay 1-year LIBOR − 25 bps for the remaining two years, or should it continue to pay $x\%$ for the remaining two years? Explain.

d) This question is a continuation of part c) above. Suppose again that two years have gone by. Swaps 'R Us has just decided whether or not to exercise the Flip-Flop Option and has informed the client of its decision. Based on the rational exercise decision of Swaps 'R Us, what is the mark-to-market value of the swap?

e) This question is a continuation of part c) above. Suppose again that two years have gone by. Suppose that, due to a clerical error, Swaps 'R Us incorrectly determines whether or not to exercise the Flip-Flop Option (i.e., if Swaps 'R Us should exercise, it doesn't, and if it shouldn't exercise, it does). Suppose Swaps 'R Us informs the client of its (incorrect) decision. Based on this incorrect decision, what is the mark-to-market of the swap?

f) For this question, ignore all parts above. Suppose that the Flip-Flop Option given above in the original terms of the trade is modified as follows:

Flip-Flop Option:	After two years, *and each year thereafter*, Swaps 'R Us has the right to pay 1-year LIBOR − 25 bps for that year

Thus, in this trade Swaps 'R Us can decide after two years if it wants to pay a fixed coupon or 1-year LIBOR − 25 bps in the third year. And irrespective of what Swaps' R Us decides to do at the end of year two, it can decide after three years if it wants to pay a fixed coupon or 1-year LIBOR − 25 bps in the final year. Denote the fixed coupon in this trade by y. What is y?

g) How does the fixed rate y in the trade you just solved for compare with the fixed rate x that you solved for in part a) above? Explain.

h) This question is a continuation of part f) above. Suppose that two years have gone by. Today is the day that Swaps 'R Us must decide if it wants to pay a fixed coupons of $y\%$ at the end of this year or if it instead wants to pay floating coupons equal to 1-year LIBOR − 25 bps at the end of this year. (Again, irrespective of the decision made by Swaps 'R Us today, it will again decide in one year whether or not to pay a fixed coupon of $y\%$ in the final year or to instead pay 1-year LIBOR − 25 bps in the final year.) Should Swaps 'R Us exercise its option and pay 1-year LIBOR − 25 bps this year, or should it pay $y\%$ this year? Explain.

i) This question is a continuation of part h) above. Suppose again that two years have gone by. Swaps 'R Us has just decided whether or not to exercise the revised Flip-Flop Option presented in part f) and has informed the client of its decision. Based on the rational exercise decision of Swaps 'R Us, what is the mark-to-market value of the swap?

12. Suppose that today's swap curve is as follows (these rates annual bond and assume that the floating side is 1-year LIBOR paid annual bond as well):

Maturity	Swap Rate
1 year	5.25%
2 year	5.50%
3 year	5.75%
4 year	6.00%
5 year	6.25%

A *(full) knockout swap* is a fixed-floating swap in which payments on each side can be "knocked out" entirely (meaning *all* subsequent cash flows will no longer be made) if some reference rate reaches a certain level. To take a concrete example, consider a $100mm 5-year swap in which a client receives fixed annual bond and pays 1-year LIBOR to the swap dealer Swaps 'R Us.

This swap has the following knockout feature: if, at the end of one year, the then-prevailing 4-year swap rate is less than or equal to the *knockout trigger* of 5.75%, then none of the fixed or the floating payments for the remainder of the swap will be made. That is, if in one year's time 4-year swap rates are less than or equal to 5.75%, then payments will be made in the swap at the end of year one, and none of the remaining cash flows in the remaining four years will occur.

To summarize, the structure of the swap is as follows:

Swaps 'R Us Pays:	Fixed
Client Pays:	1-year LIBOR
Notional:	$100mm
Maturity:	5 years
Payment Frequency:	Annual, on both sides
Day Count:	30/360, on both sides
Knockout Feature:	If, in one year, the 4-year swap rate is less than or equal to the Knockout Trigger, then there will be no further payments in the swap on either side
Knockout Trigger:	5.75%

Furthermore, assume that the base vol curve is flat at 18% and suppose that the swaption matrix is flat at 20%.

a) What is the fixed coupon in the swap? That is, what fixed coupon gives this swap an NPV = 0?

b) Consider now changing the Knockout Trigger and recalculating the fixed coupon in the knockout swap. Provide a graph that shows how the fixed coupon in the knockout swap varies with the Knockout Trigger for a reasonable range of Knockout Triggers. Interpret your findings.

13. Suppose the swap curve is as follows:

Maturity	Swap Rate
1 year	5.60%
2 year	6.00%
3 year	6.20%
4 year	6.40%
5 year	6.60%

All swap rates assume annual bond fixed payments versus 1-year LIBOR.

a) Consider a 4-year swap with LIBOR set in arrears. What is the 4-year swap rate versus LIBOR set in arrears (in computing this rate, ignore any convexity adjustments)?

b) How does your answer in a) compare to the 4-year swap rate with LIBOR set the normal way (i.e., set in advanced and paid in arrears)?

c) If we incorporate the convexity adjustment, will the fixed rate be higher or lower than the fixed rate you solved for in a)?

14. Consider our stylized example and the following trade:

Swaps 'R Us Pays:	$X\%$
Client Pays:	5-year CMS
Notional:	\$100mm
Maturity:	1 year
Floating Reset:	One time, in one year
Fixed Payment:	Fixed $\times$ Notional $\times$ 360/360
Floating Payment:	5-year CMS $\times$ Notional $\times$ 360/360
Payment Date:	One year

Suppose further that the 5-year swap rate, one year forward is 6%.

a) Which of the following is true of the fixed rate X?

(1) X is greater than 6%.

(2) X is less than 6%.

(3) X is exactly 6%.

(4) There is not enough information to determine conclusively.

b) Suppose that just prior to executing this trade the level of all implied swaption volatilities increases. Assume there are no additional changes in the market. What happens to the level of X that Swaps 'R Us is willing to pay in the trade?

(1) X increases.

(2) X decreases.

(3) X does not change at all.

(4) There is not enough information to determine conclusively.

Chapter 7

Structured Notes

> It was the best of times, it was the worst of times, it was the age of wisdom, it was the age of foolishness, it was the epoch of belief, it was the epoch of incredulity, it was the season of Light, it was the season of Darkness, it was the spring of hope, it was the winter of despair, we had everything before us, we had nothing before us, we were all going direct to Heaven, we were all going direct the other way.[1]

In the early 1990s, the U.S. economy was in a deep recession, and the Fed had cut short-term interest rates significantly in an effort to stimulate the economy. Investors who had previously enjoyed a period of high interest rates and high returns from the fixed income markets were now faced with the prospect of historic low yields. At the same time, the sophistication of the derivative markets was on the rise, allowing dealers to price and hedge more complex, riskier trades. Such trades would often offer potentially high returns to investors but came at the expense of tremendous risks—risks that were not necessarily completely understood.

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[1]From *A Tale of Two Cities* by Charles Dickens, originally published in weekly installments in his own periodical *All the Year Round* from April 30 to November 29, 1859. Although Dickens was writing about a time during the French Revolution, these words are not a bad summary of the early 1990s that saw tremendous growth and innovation (and some abuses) in the structured note market, leading up to the ultimate bankruptcy of the Orange County investment portfolio as a result of its investment in highly complex structured notes.

Structured notes (SNs) represented the marriage of traditional fixed income debentures and derivatives. Prior to the advent of structured notes, the coupons paid by bonds were typically either fixed rate or floating rate in nature, and there was not too much possible variation otherwise.[2] In structured notes, an underlying derivatives transaction is embedded in the bond, allowing for a coupon type that is as nontraditional as the dealer community can create or investors can demand.[3] A myriad of products were created rapidly in the mid-1990s, some of which had disastrous consequences for the investors who bought them.[4]

After Orange County declared bankruptcy in late 1994 as a result of its investment in highly leveraged structured notes, the structured note market came to a nearly complete halt.[5] Few would buy structured notes afterward.

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[2]While it is true that the advent of the collateralized mortgage obligation (CMO) market in the early 1980s allowed for a variety of coupon types not otherwise seen (e.g., inverse floaters), the range of coupon types available in the CMO market was relatively small as compared to the structured note market. Traditional debentures issued by banks, corporations, or the U.S. agencies did not offer such variety.

[3]Some structured notes also contain underlying principal-linked derivatives, allowing investors the opportunity to obtain principal in excess of the face amount of the bond in certain states of the world.

[4]This description of structured notes (correctly) suggests that a structured note is often just a swap with embedded options that is then embedded in a bond. Historically certain swaps with embedded options have more commonly appealed to investors in *funded form* (meaning through structured notes) and are included in this chapter. Other such swaps have typically appealed to investors in *unfunded form* (meaning through the swap itself) and were discussed in Chapter 6. And some trades have been popular in both funded and unfunded form.

Given the work we did in Chapter 6, why then would investors not just always enter into the underlying swap with embedded options? Why would they want to instead buy a structured note that has the same swap embedded that they could otherwise trade directly with a dealer? There are at least two answers to these questions. First, some investors who manage portfolios simply need to invest cash. Although entering into a swap with embedded options may provide them with the exposure that they are looking for, doing so does not require an investment of cash. Buying a structured note requires an outlay of cash. Second, in addition to the myriad of products that were created rapidly during this time, a slew of investors new to the derivatives markets rapidly entered the scene during this time. Some investors, such as money market funds, who could never traditionally "do derivatives" due to restrictions in their bylaws, could get around these restrictions by buying structured notes. Their bylaws typically had little in the way of provisions preventing them from buying many structured notes but would merely provide guidelines on maturity, credit rating, and so on. Ironically, this often meant that the notes that these new entrants to the derivatives market bought were giving them exposure to some of the most complex and risky derivatives trades created.

[5]This is not to suggest that Orange County only invested in structured notes, which certainly was not the case. More important, as we will discuss later in this chapter, the real culprit behind the bankruptcy of Orange County was not so much its investment in structured notes but rather its use of repurchase agreements that allowed it to lever up and buy even more notes. In the end it was Orange County's inability to fulfill its obligations under its repos that caused its bankruptcy and ultimate liquidation of its investment portfolio.

But time heals all wounds. After months, and in some cases, years, investors returned from the sidelines and bought structured notes again. Many new investors have entered the structured note market over the years.

It is important to bear in mind that we will be looking only at one segment of the structured note market: interest rate–linked structured notes. There are structured notes created that give investors exposure to a variety of other asset classes, including credit, commodities, and equities. In this chapter we will examine a variety of structured notes that are common in the market. We'll take a look at how some of these notes are priced. And we will examine the benefits and risks inherent in these bonds. For a more thorough examination of the structured note market, see, for example, Peng and Dattatreya (1995) or Knop (2002).

7.1 The Rise of the Structured Note Market

The structured note market first came into the spotlight in the early 1990s. Peng and Dattatreya (1995) report that volume of newly issued structured notes increased more than fourfold from 1990 to 1994, with structured note new issue volume at under \$20bn in 1990 and in excess of \$80bn in 1994. There were two factors in the market in the early 1990s that contributed to such significant growth. First, the U.S. economy had plunged into recession in the early 1990s, and in an effort to stimulate the economy the Fed cut the Federal Funds rate significantly during this time. The funds rate declined from a high of 8.25% in 1990 down to 3% in 1992. Investors who had enjoyed high yields (and high returns) in the fixed income markets were now faced with the prospect of a low rate environment. Of course for many investors, particularly money market investors, LIBOR is more of a benchmark than is the Fed Funds rate. Figure 7.1 shows a history of 3-month LIBOR during this time. In the early 1990s, many investors were starving for the high yield levels that they had enjoyed only a few years before and were looking for alternative investments that would bring them high returns.

At the same time, the innovation found in the derivatives markets was rising dramatically. New types of structured notes were being created that provided investors with high yields. In order to create such high (current) yields for investors, such notes often required their holders to assume tremendous risks. Leveraged inverse floaters and range accrual notes (both of which we shall look at) are examples of such notes. These notes in general performed very well in the early 1990s. But by the end of 1994, a year during which the Fed had raised the Fed Funds rate six times, the results for investors who held such bonds were often disastrous. As shown in Figure 7.1, during this period 3-month LIBOR increased dramatically, hurting investors who had expressed the view that LIBOR would remain low.

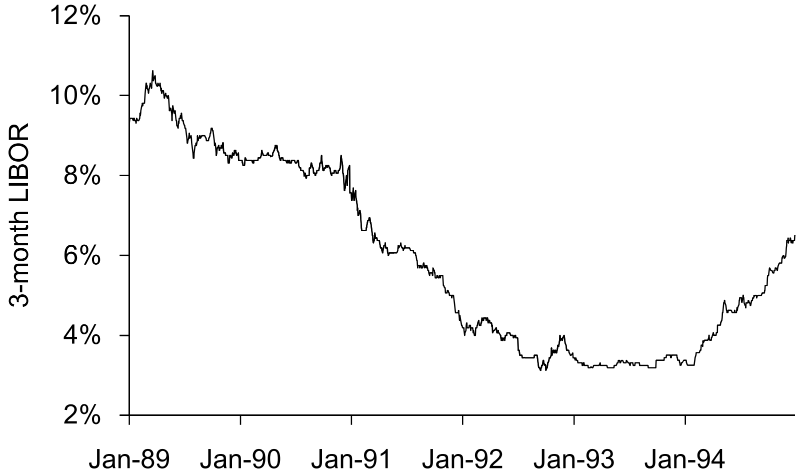

Figure 7.1: Historical 3-Month LIBOR

7.2 A Glossary of Structured Notes

It's impractical to list every type of structured note out there, because these notes are typically customized for the investors who buy them. Nevertheless, there are a number of basic types of structured notes that have been common over the years, some of which remain popular with investors today. Before beginning to analyze structured notes in detail, in this section we will present a menu of various types of structured notes that have been common in the market. As previously mentioned, although the structured note market rose from the interest rate derivatives market, over the years notes have been created that give investors exposure to a variety of assets classes, including equities and commodities. We will focus exclusively on structured notes tied to interest rate derivatives.

Structured notes are characterized first and foremost by the type of coupons they bear. The following list, while not complete, contains a number of varieties:

- **Floater:** A bond whose coupon depends on a floating rate index, such as LIBOR, Prime, T-Bills, or Fed Funds. As an example, such a bond may pay the following coupons:

$$\text{Years 1–5:} \quad \text{Prime} - 2.90\%$$

- **Deleveraged Floater:** A bond whose coupon depends on a percentage of the movement in some underlying floating rate index. Often these bonds initially pay an (above market) fixed rate coupon for some period of time, followed by floating coupons. As an example, such a bond may pay the following coupons:

Year 1:	5%
Years 2–5:	$0.5 \times$ 3-month LIBOR + 1.50%

- **Capped/Floored Floater:** A bond that pays a floating coupon, which has a stated maximum or minimum coupon level, regardless of where the underlying floating rate sets. These bonds may also be callable. As an example, such a bond may pay the following coupons:

Years 1–5:	6-month LIBOR + 30 bps, subject to a maximum coupon of 8%

  Other variations of this bond allow for the maximum or minimum coupon to vary over the life of the bond. These bonds are referred to as *stepped capped floaters* or *stepped floored floaters*.

- **Callable Multi-Step Bonds:** A bond that pays a "teaser" rate of interest, higher than a comparable fixed rate, bullet maturity bond, and then is either called or "steps up" to a higher coupon rate. Such a bond may pay successively higher fixed rate coupons in later periods of its life, assuming the bond has not yet been called (and is referred to as a *variable step-up bond*).

Year 1:	6%
Year 2:	7%
Year 3:	8%
Year 4:	10%
Year 5:	12%
Call Feature:	This bond is callable in one year and every quarter thereafter

- **Extendible:** A bond that often pays a "teaser" rate of interest, higher than a comparable fixed rate, bullet maturity bond, and the issuer has the right to extend the bond for an additional period of time.

Year 1:	6%
Extend Feature:	This issuer has the right in one year to extend the maturity of the bond for an additional four years

- **Flipper:** A bond whose coupon can "flip" from fixed to floating after a period of time, at the option of the issuer.[6] As an example, such a bond may pay the following coupons:

  Year 1: 6%
  Years 2–5: 6% or 3-month LIBOR − 25 bps, at the option of the issuer

- **Index Amortization Notes (IANs):** A bond that may repay principal prior to the stated maturity based on a predetermined amortization schedule and change in a reference rate. As an example, such a bond may pay the following coupons:

  Years 1–5: 6%, but if 5-year CMT $<$ 5% on any reset date, then the entire bond is retired early

- **Inverse Floater:** A bond whose coupon varies inversely with some floating rate of interest. As an example, such a bond may pay the following coupons:

  Years 1–5: 6% − 3-month LIBOR, subject to a minimum coupon of 0%

- **Leveraged Inverse Floater:** A bond whose coupon varies inversely with some floating rate of interest. However, a multiplier that is greater than one is used in the coupon formula to amplify movements in the coupons given movements in the underlying floating rate of interest. As an example, such a bond may pay the following coupons:

  Years 1–5: 15% − 4 $\times$ 3-month LIBOR, subject to a minimum coupon of 0%

- **Dual-Index Floater:** A bond whose coupon depends on two different floating rate indexes. Often these bonds initially pay an (above market) fixed rate coupon for some period of time, followed by dual-index floating coupons. As an example, such a bond may pay the following coupons:

  Year 1: 5%
  Years 2–5: 10-year CMS − 3-month LIBOR + 0.50%, subject to a minimum coupon of 0%

[6] Question 11 of Problem Set 6 gave you an opportunity to look at the underlying swap used to create such a structured note.